Type MS · Estimate an unmeasured value by interpolating or extrapolating data

Answer 1 of 4 · Question 4 part b · 4 marks

2025 JUN · Paper 02 · extract 1 of 2 · find part b in this mark scheme extract

4 (a) (i)		<table><tr><th>Conc (sugar) in au</th><th>Percentage change in mass</th></tr><tr><td>0.00</td><td>8.0</td></tr><tr><td>0.25</td><td>4.0</td></tr><tr><td>0.50</td><td>-1.0</td></tr><tr><td>1.00</td><td>-7.0</td></tr><tr><td>1.50</td><td>-10.0</td></tr></table> • two row/column table; • concentration (sugar) in au; • percentage/% change in mass; • 0.0 to 1.50/1.50-0.00;	Ignore units in the body of the table Accept AU for MP2 Ignore % symbol for MP3 mp3 unit not required for percentage in mass	4
	(ii)	0.50 (au);		1
	(iii)	$\frac{104 \times 7}{100}$; = 7.28 g; =7.3;	Allow 4/100 for 1 mark or 0.28	3
(b)		• change in mass is -1% to +4% • 5% range in mass • between 0.25 and 0.5 au; • accept any answer between 0.375- 0.5; ***an answer of o.45 scores =4 marks***	MP 1-4 are separate MPs unless the answer is 0.45	4
(c) (i)		• dry the cylinder/ remove water/liquid; • affects mass/wrong mass/heavier/inaccurate mass/eq;	Accept weight	2
	(ii)	any two from • volume of (sugar) solution; • temperature; • surface area of cylinders; • same type of potato;	Ignore amount accept volume of liquid	2
				Total =16

Type MS · Estimate an unmeasured value by interpolating or extrapolating data

Answer 1 of 4 · Question 4 part b · 4 marks

2025 JUN · Paper 02 · extract 2 of 2 · find part b in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Estimate an unmeasured value by interpolating or extrapolating data

Answer 2 of 4 · Question 4 part b.i · 1 mark

2022 JUN · Paper 01 (regional R) · extract 1 of 2 · find part b.i in this mark scheme extract

4 (a) (i)	Any three of the following: • add alcohol/ethanol to food; • shake; • add water; • milky coloration shows fat present;	Allow rub food; onto filter paper; translucent; answer must be in sequence i.e. ethanol first	3
(ii)	lipase		1
(iii)	fatty acids; glycerol;		2
(b) (i)	Any value between 8.6 and 8.8;		1
(ii)	independent = time/ temperature; dependent = pH;		2
(iii)	• 37°C is the optimum temperature; • lipase/enzyme is more active/works faster; • fat is digested more quickly; • at 80 °C/higher temperature enzyme becomes denatured;	Allow reverse argument throughout: • at 80°C/higher temperature enzyme is denatured; • enzyme less active; • fat digested more slowly; • 37°C is the optimum temperature (for enzyme activity/enzyme works faster at 37°C	4

Total marks 13

Type MS · Estimate an unmeasured value by interpolating or extrapolating data

Answer 2 of 4 · Question 4 part b.i · 1 mark

2022 JUN · Paper 01 (regional R) · extract 2 of 2 · find part b.i in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Estimate an unmeasured value by interpolating or extrapolating data

Answer 3 of 4 · Question 5 part e · 1 mark

2021 NOV · Paper 02 · extract 1 of 2 · find part e in this mark scheme extract

number			
5 (a)	• addition of figures = 1690; • divide by 10 = 169;		2
(b)	makes it faster/decreases reaction time;		1
(c)	• caffeine has to be absorbed; • into blood; • transported to nervous system/brain;	Ignore body	3
(d)	• can't control what is in food/drink; • could contain caffeine/other factors that would affect result/to ensure caffeine is the cause;	Ignore coffee	2
(e)	185 ± 5 mm;		1 Total 9

Type MS · Estimate an unmeasured value by interpolating or extrapolating data

Answer 3 of 4 · Question 5 part e · 1 mark

2021 NOV · Paper 02 · extract 2 of 2 · find part e in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Estimate an unmeasured value by interpolating or extrapolating data

Answer 4 of 4 · Question 7 part a.iii · 2 marks

2023 JUN · Paper 01 · extract 1 of 2 · find part a.iii in this mark scheme extract

Number				
7 (a)	(i)	• axes labelled with units; • correct way round and suitable scale; • correct plots; • suitable line drawn;		4
	(ii)	• only not working correctly when rate falls below 100; • data/year not shown when that occurs/already below 100;		2
	(iii)	• 2021/2022; • rate falling at 5-10 units per annum/similar values each year;		2
	(iv)	• not very accurate/results vary/fluctuate; • earlier data showed improvement in function; • could happen again/results may change;		3
(b)		any three from • protein broken down into urea/deamination; • toxic; • needs to be excreted; • incorrectly functioning kidney can't remove it all if there is a lot; • less protein less urea;		3

Total 14

Type MS · Estimate an unmeasured value by interpolating or extrapolating data

Answer 4 of 4 · Question 7 part a.iii · 2 marks

2023 JUN · Paper 01 · extract 2 of 2 · find part a.iii in this mark scheme extract

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